

EEE 312 – Digital Signal Processing I Laboratory
Jan 2026 Level-3 Term-I Section C2
Final Project Demonstration

Detecting Alzheimer's Disease from Resting-State EEG

A validated digital signal processing pipeline, tested on people it has never seen

SUBMITTED BY – GROUP 01



Shahriar Nafis Rabbi
2206184



H M Hamim Al Muhit
2206169



Md. Ashique Al Arafat Nayeem
2206190



Arib Salehin
2206186



BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Shahriar Nafis Rabbi
2206184

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H M Hamim Al Muhit
2206169

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Md. Ashique Al Arafat Nayeem
2206190

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Arib Salehin
2206186

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Live demo in the Q&A slot: one recording in → its spectrum on screen → probability of Alzheimer's out.



Shown for reference, not presented: 6.1 Contributions · 6.2 Logbook · 8 References



1. Summary / Abstract

We designed, implemented and verified the complete DSP chain that turns a resting, eyes-closed EEG into a probability of Alzheimer's disease, and tested it honestly on people the model had never seen.



0.81

AUC, AD vs healthy, people never seen
95 % CI 0.72–0.90

$p = 0.005$

permutation test
no label shuffle of 200 reached it

7 numbers

regional θ/α ratios alone
AUC 0.83, as good as all 94

5 / 5

robustness criteria met
fixed before the runs

2. Introduction



Alzheimer's disease

The most common cause of dementia (60–70 % of cases, WHO). Early detection gives time to treat and plan.



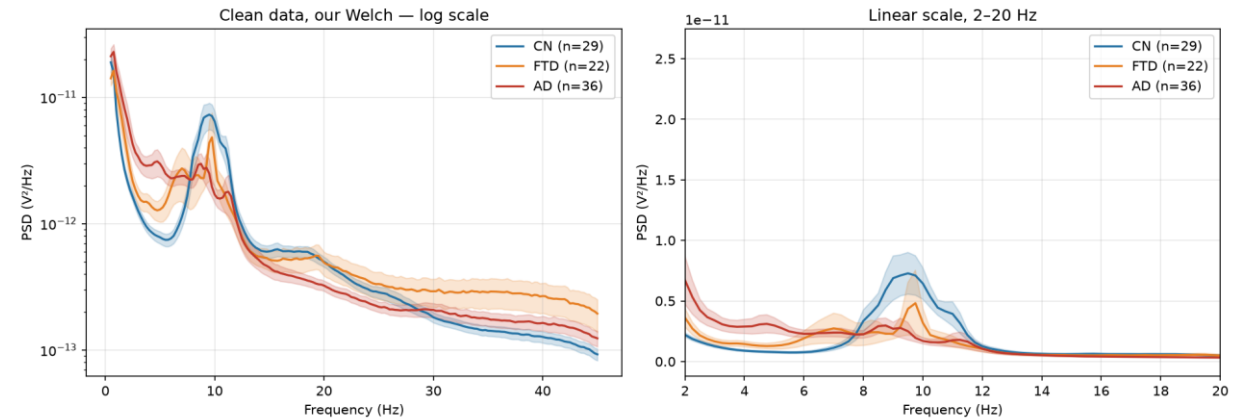
Today's tests are costly

MRI, PET and spinal-fluid tests are expensive and scarce outside large hospitals.

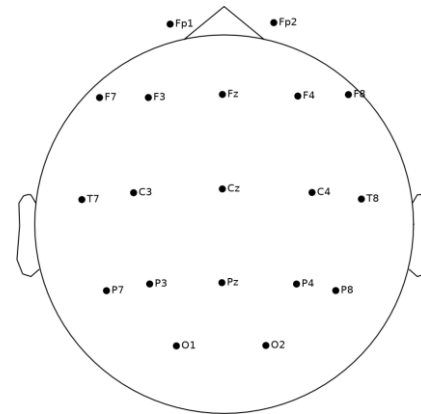


EEG is cheap, portable, harmless

And in AD the brain rhythm slows: alpha (8–13 Hz) weakens, theta (4–8 Hz) grows.



Grand-average spectra after our pipeline: the ~10 Hz alpha peak is weaker and slower in AD.



Dataset: OpenNeuro ds004504

- 88 people: 36 AD · 23 FTD · 29 healthy (CN)
- 19 scalp electrodes (10–20), 500 Hz
- Resting state, eyes closed
- 1 FTD excluded (under 2 min clean) → 87

3.1 Design: Problem Formulation (PO(b))

3.1.1 Identification of Scope

In scope

- ✓ Resting, eyes-closed EEG (ds004504)
- ✓ AD vs healthy (main); FTD as a second question
- ✓ Every DSP step designed, coded and verified by us
- ✓ Honest testing on people never seen

Out of scope

- ✗ A clinical diagnosis (screening support only)
- ✗ Collecting new recordings
- ✗ Hardware or a real-time device
- ✗ Deep learning (only 65 AD + CN people)

3.1.2 Literature Review

1

EEG slowing in AD

Alpha ↓ and theta/delta ↑ is the most replicated EEG change in Alzheimer's (Jeong, 2004).

2

Our dataset

Miltiadous et al. (2023): 88 recordings of AD, FTD and healthy people, with a subject-wise baseline.

3

A common pitfall

Many studies report 95–99 % by putting epochs of the same person in training and test (data leakage; Varoquaux et al., 2017).

4

The gap we address

A verified DSP chain, a result declared in advance, and ablation and robustness tests.

3.1 Design: Problem Formulation (PO(b))

3.1.3 Formulation of Problem

Input $x[n] \in \mathbb{R}^{19}$, $f_s = 500$ Hz, eyes closed

Output $P(\text{AD} \mid x) \in [0, 1]$

Goal maximise the AUC on people never used for training

Declared before any result AD vs CN · 94 regional features · logistic regression · seed 2026

Constraints

- No information from a test person may reach training (nested cross-validation)
- Every stage must be checkable: built-in checkpoints

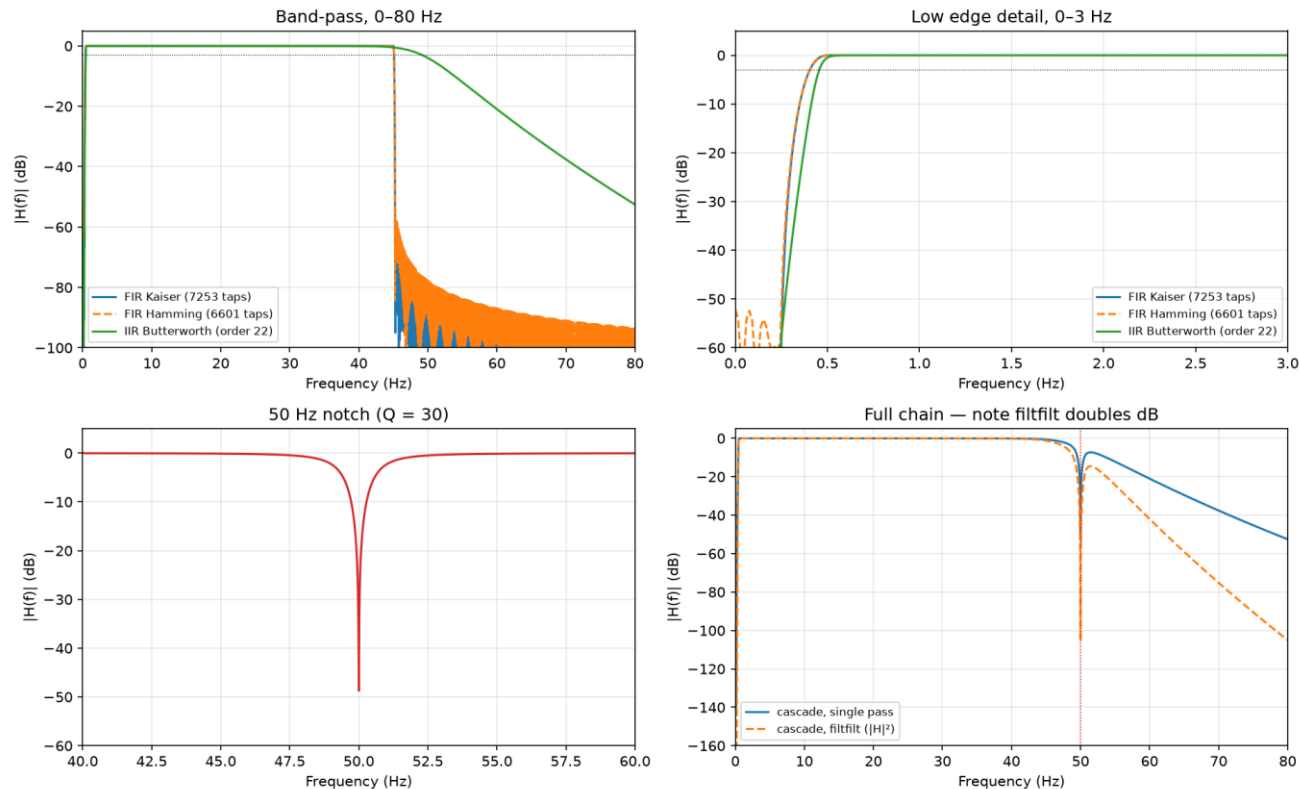
3.1.4 Analysis: signal properties → design specifications

Signal property	Requirement	Our design
Brain rhythms lie in 0.5–45 Hz	remove drift and muscle noise	band-pass 0.5–45 Hz
50 Hz mains interference	≥ 40 dB rejection	notch, $Q = 30$
Band ratios compare timing	no phase distortion	zero phase (forward–backward)
Content < 45 Hz at 500 Hz	keep Nyquist margin	resample to 250 Hz
θ (4–8) vs α (8–13) peaks	resolution ≤ 0.25 Hz	4 s Welch segments
Blinks of hundreds of μV	remove, keep the data	ICA + 150 μV rejection



3.2 Design Methods (PO(a))

Filter design: band-pass 0.5–45 Hz and a 50 Hz notch, both zero phase



Magnitude responses of our designs (full band, low edge, notch, full chain).



FIR reference design

Kaiser window, 7253 taps, transition 0.25 Hz. Linear phase but a 7.25 s delay.



IIR design (used)

Butterworth band-pass, order 22 = 11 second-order sections. $|H(j\Omega)|^2 = 1 / (1 + (\Omega/\Omega_c)^{2N})$



50 Hz notch

Q = 30. Whole chain: -52.7 dB at 50 Hz in one pass, doubled by forward-backward filtering.

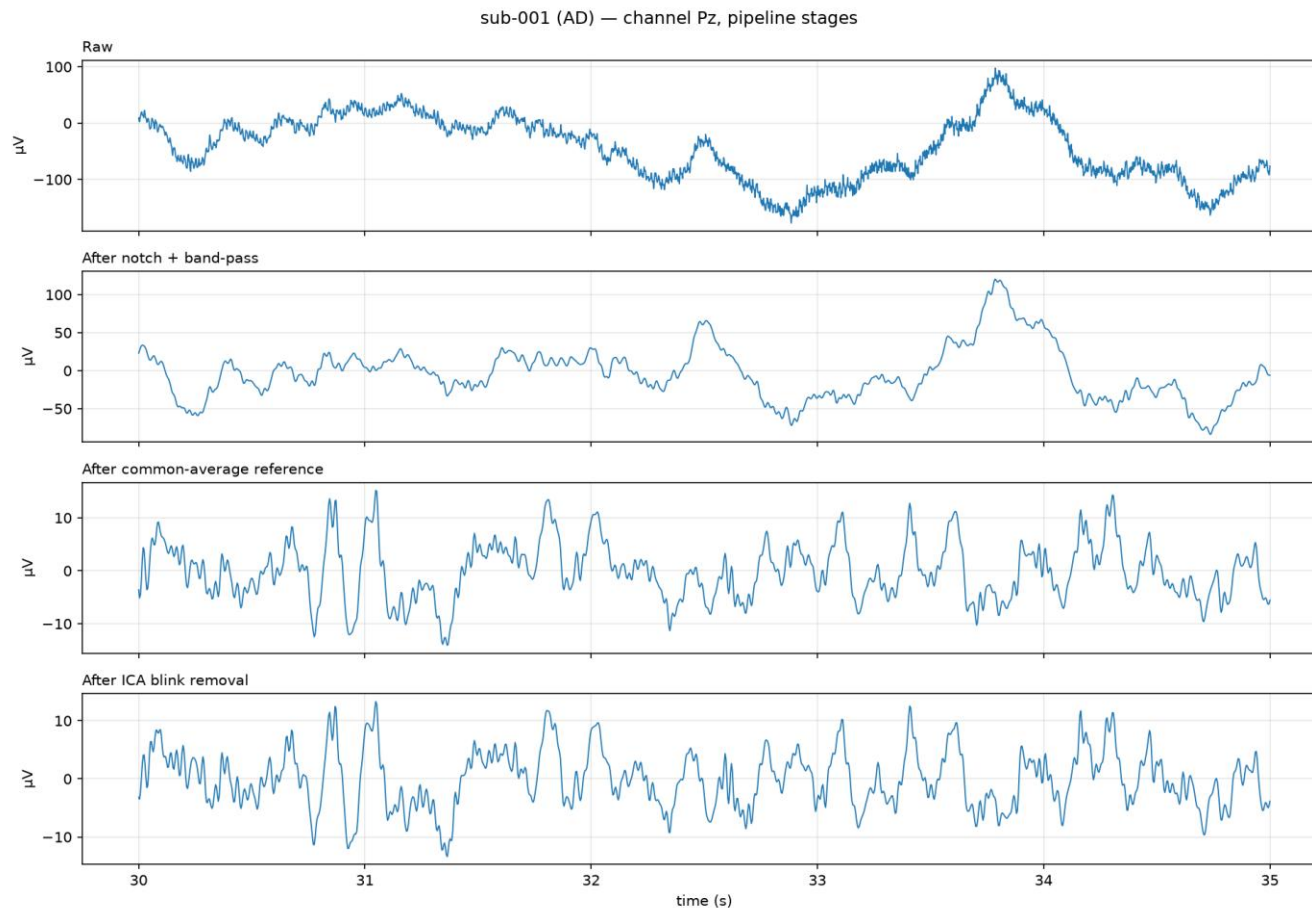


Zero phase

Forward-backward filtering (sosfiltfilt): the phase cancels, $|H|^2$ remains. No timing distortion.

3.2 Design Methods (PO(a))

Removing what is not brain: re-referencing, ICA and epoch rejection



One AD patient, channel Pz, after each stage of our pipeline.

- 1 Crop 10 s at each end (filter edge transients)
- 2 Resample 500 $\rightarrow$ 250 Hz (Nyquist 125 Hz $\gg$ 45 Hz)
- 3 Rebuild one faulty electrode (sub-067, T8)
- 4 Common-average reference
- 5 ICA, 15 components: remove the blink component
- 6 4 s epochs, reject any > 150 μ V peak-to-peak

ICA makes data quality equal

Epochs lost to the 150 μ V rule:

without ICA AD 14 % vs CN 8.7 %

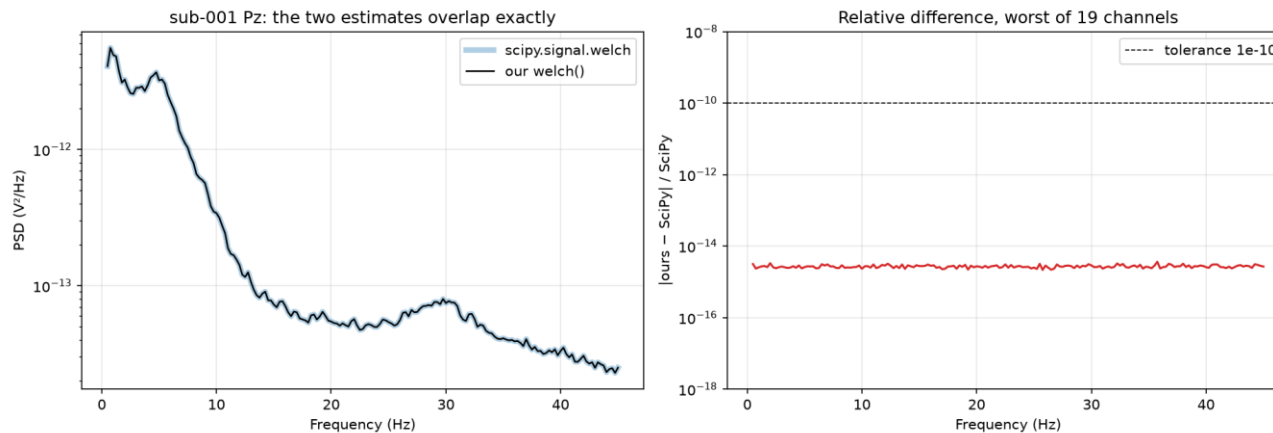
with ICA AD 5.3 % vs CN 5.6 %

3.2 Design Methods (PO(a))

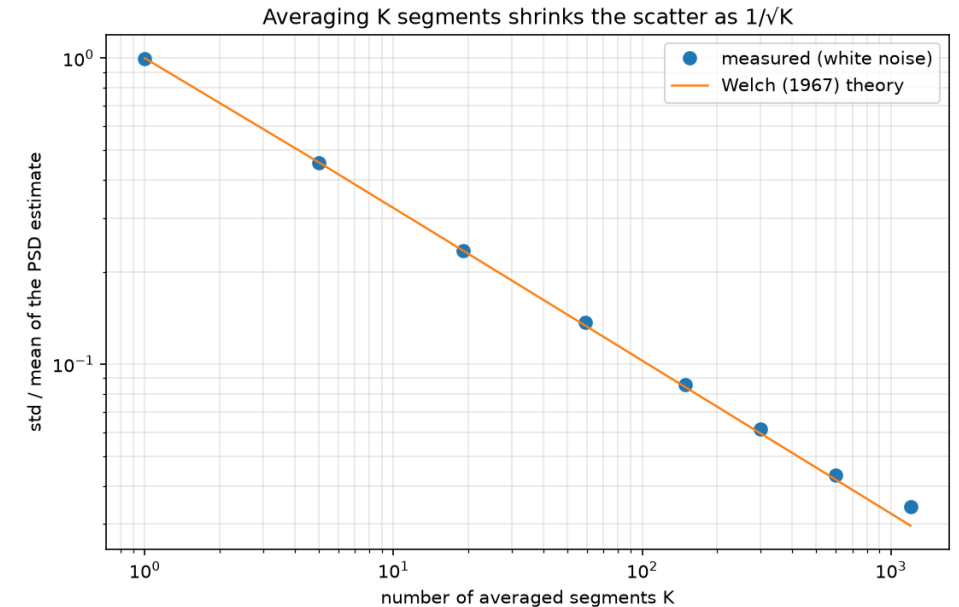
Spectral estimation: our own Welch estimator (NumPy), validated against SciPy

$$P(f) = \frac{1}{K} \sum_k \frac{\left| \sum_n w[n] x_k[n] e^{-j2\pi f n} \right|^2}{f_s \sum_n w^2[n]}$$

- Hann window, $T = 4$ s $\rightarrow \Delta f = 1/T = 0.25$ Hz
- 50 % overlap, $K \approx 370$ clean segments per person
- Variance falls as $1/K$ (Welch, 1967): verified on white noise



Left: our estimate and SciPy's overlap. Right: relative difference, far below the 10^{-10} tolerance.



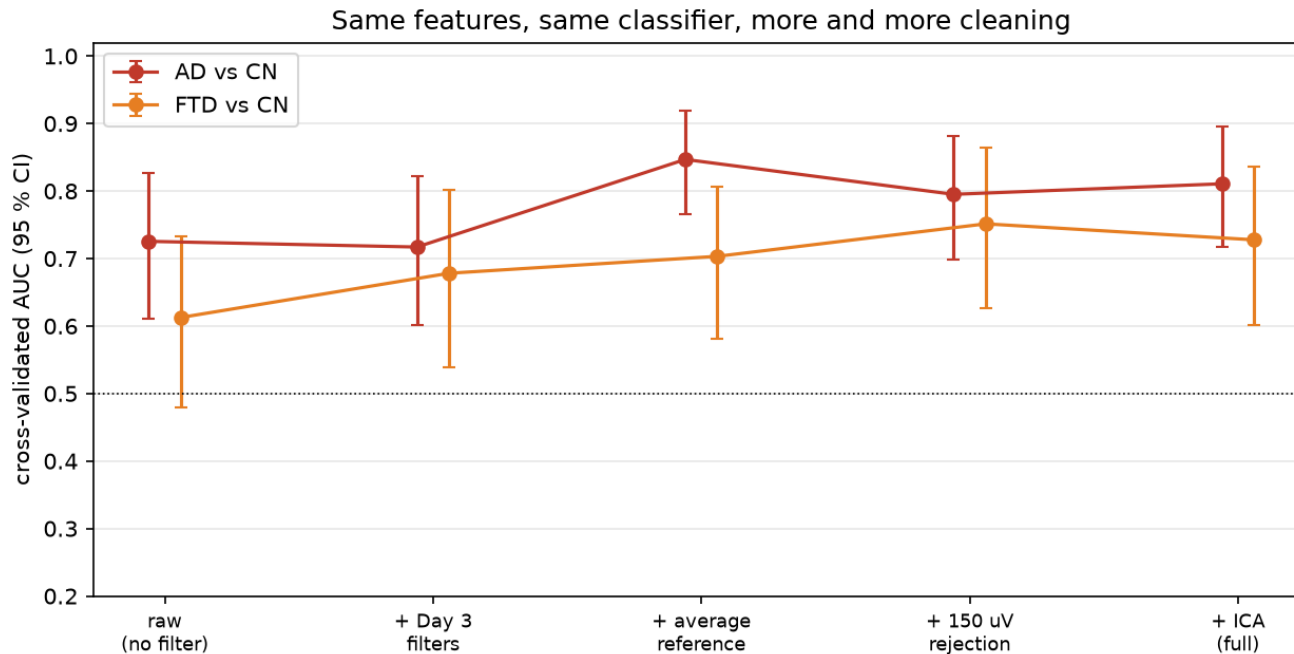
Why it matters

One periodogram has a standard deviation equal to its mean. Averaging 370 segments shrinks it about 19×, so band powers become stable numbers.



3.3 Design: Simulation — Does Each DSP Step Matter?

Cleaning ablation: the same 94 features and the same classifier, with more and more DSP



Key step: the average reference (+0.13 AUC). The ear reference adds one shared rhythm to every channel and makes all regions look alike.

Level	AD vs CN	FTD vs CN
Raw (no filter)	0.725	0.613
+ our filters	0.717	0.678
+ average reference	0.847	0.703
+ 150 μ V rejection	0.795	0.751
+ ICA (our pipeline)	0.811	0.728

Why report 0.811, not 0.847?

Eye movements add slow δ/θ power that looks like AD slowing, and patients had more of them. A higher AUC for the wrong reason is not a result.



4.1 Implementation: Features and Honest Validation

94 features = 13 measures × 7 regions + 3 alpha-peak

Band powers **35**

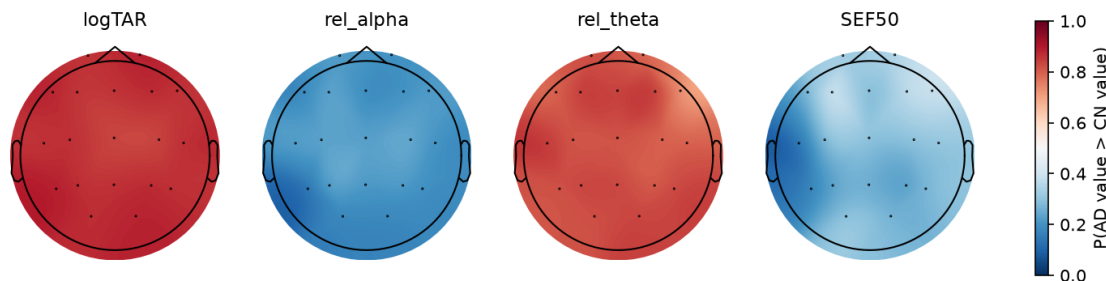
θ/α , slow/fast ratios **14**

Spectral shape **21**

Hjorth **21**

Alpha peak **3**

Where on the scalp each feature separates the groups (0.5 = no difference)



Where each feature separates AD from CN (AUC per electrode).



Leakage test on pure noise

Features chosen on everyone first: AUC 0.78 (fake). Chosen inside the folds: 0.46 (correct). So every choice happens inside training.

Outer loop — the judge: 5 folds × 10 repeats

test 13

train

train

train

train

each person is tested once per repeat, by a model that never saw them

Inner loop — the tuner: 5 folds inside the 52 training people

impute

scale

keep k best

logistic regression

chooses C (regularisation) and k (number of features)



Declared before any result

AD vs CN · region features · logistic regression · seed 2026.
Other models and comparisons are reported beside it, never swapped in.

4.2 Implementation: Demonstration

Main result: AD vs healthy, 65 people, every prediction made by a model that never saw that person

0.811

AUC
95 % CI 0.717–0.896

0.76

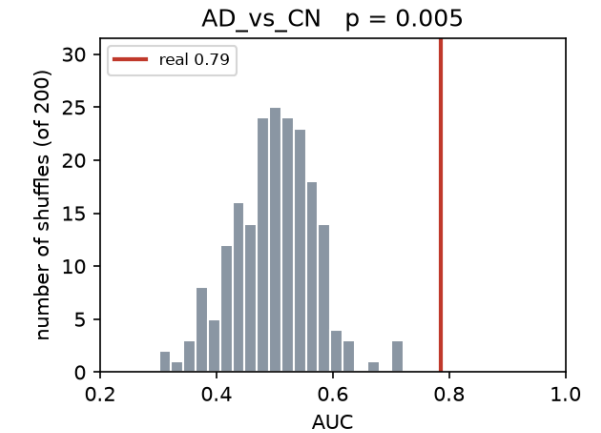
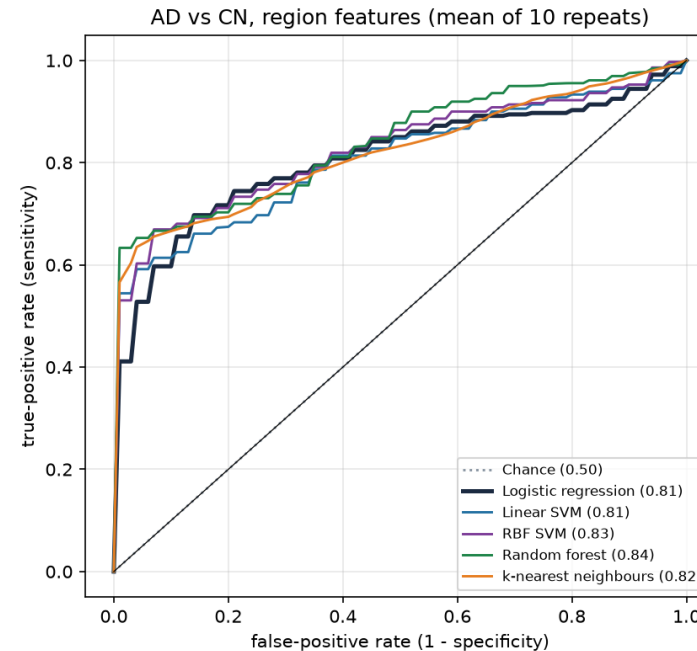
balanced accuracy
95 % CI 0.68–0.85

0.73 / 0.80

sensitivity / specificity
at probability 0.5

p = 0.005

permutation test
0 of 200 shuffles reached it



Grey: AUCs after shuffling the diagnoses. Red: the real score.

FTD vs CN
0.728 (p = 0.03)

AD vs FTD
0.578 (not separable)

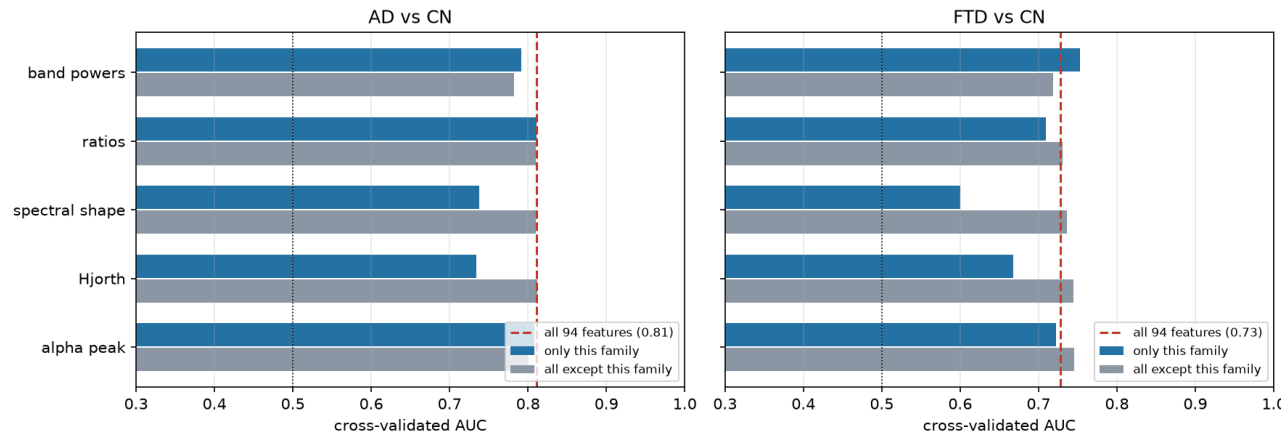
3 classes
bal. acc. 0.53 (chance 0.33)

Age + sex alone
0.624



5.1 Novelty

What drives the result?



Each feature family alone (blue) and all except it (grey); red line = all 94.

0.827

7 θ/α ratios only

0.828

16 posterior features only

0.811

all 94 features

One phenomenon, EEG slowing, carried by every family. Posterior-only rules out eye artefacts.

What is new in our work



Every DSP step our own, and verified

Welch = SciPy to machine precision; the full pipeline reruns identically (10^{-7}).



Result declared before any run

Plus a permutation test: no cherry-picking among 52 model runs.



Ablation measures each DSP step

Shows which processing actually earns its place (average reference: +0.13).



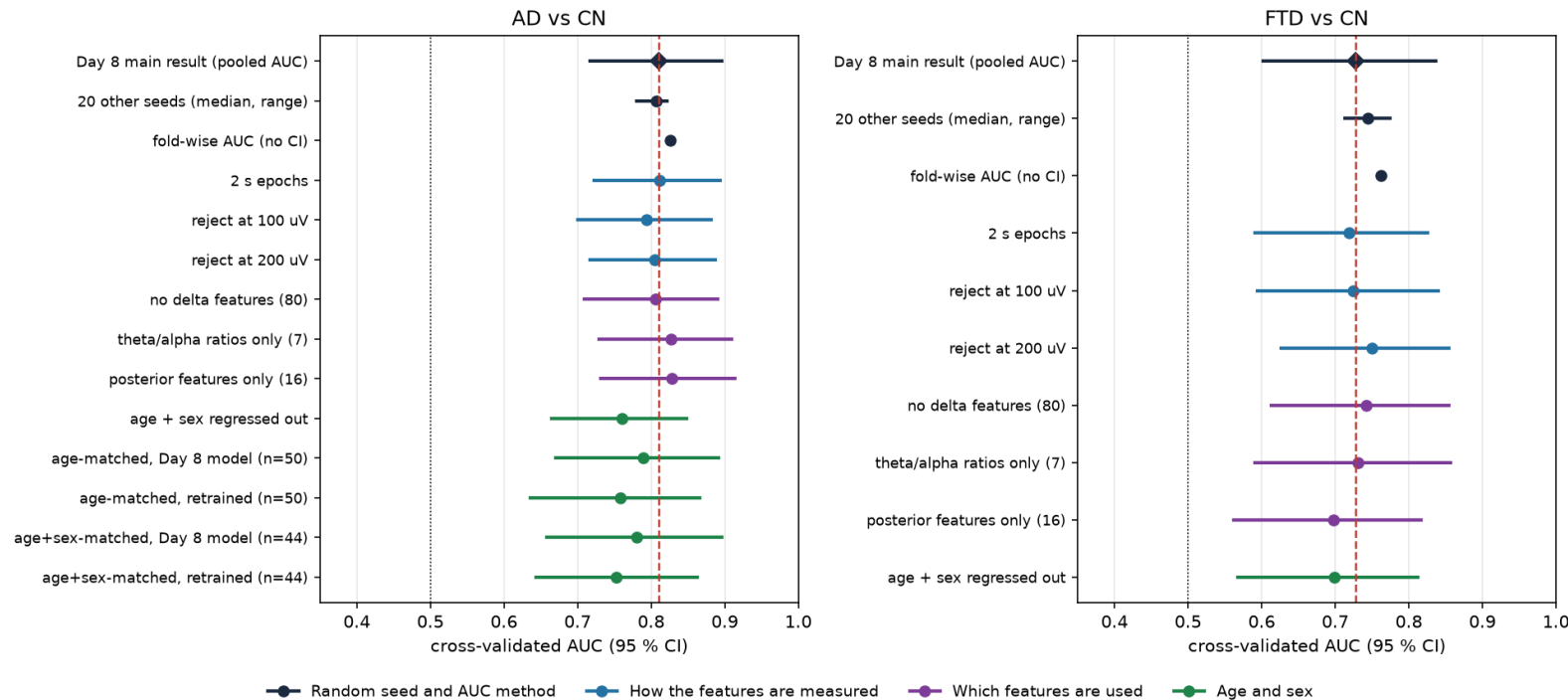
Found a flaw in public data

EEG fingerprints showed participant numbers do not match across two datasets; reported to the authors.

5.2 Investigations (PO(d))

Would other reasonable choices change the answer? Five criteria written down before the runs

Robustness of the main result: red dashed line = Day 8, dotted = chance



Every row keeps its 95 % interval far above chance (dotted line). Red dashed line = the main result.

5 / 5 criteria met

- ✓ 20 other random seeds: 0.78–0.82
- ✓ Fold-wise instead of pooled AUC: 0.826
- ✓ 2 s · 100/200 μ V · no δ : 0.79–0.81
- ✓ θ/α only 0.827 · posterior only 0.828
- ✓ EEG adds beyond age + sex: $p < 10^{-6}$

5.2 Investigations (PO(d))

Is it really the disease, not age or sex? And where do FTD patients fall?

67 % vs 38 %

women in AD vs CN ($p = 0.03$)
a real imbalance to rule out

0.781

AUC in 22 age- and sex-matched
pairs
age and sex cannot help here

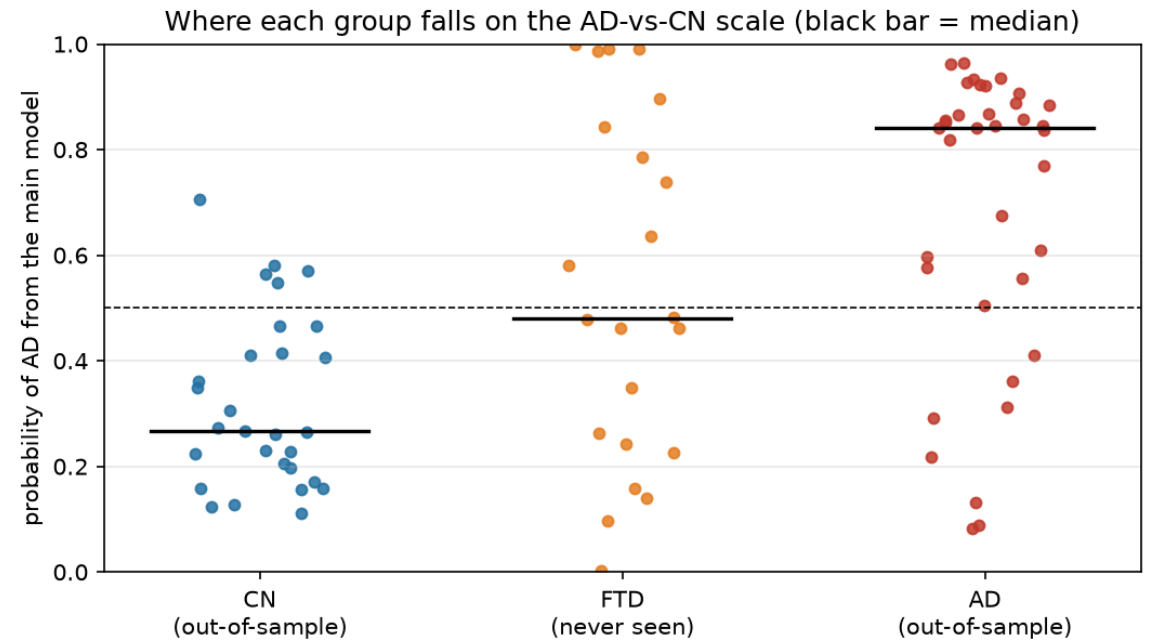
OR 5.7

per SD of the EEG score
at equal age and sex, $p < 10^{-6}$

0.761

age + sex regressed out of all
features
the only significant drop

Controlled three ways, the AUC stays 0.75–0.79: well above chance.



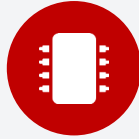
FTD patients, never seen by the model, fall between CN and AD (median 0.48): both dementias slow the EEG. That is why AD vs FTD is hard (0.58).

5.3 Limitations of Tools (PO(e))



Small, single-site sample

65 AD + CN people from one hospital: wide interval (0.72–0.90), no external test set.



Recording hardware

19 electrodes (10–20), 500 Hz, ear reference: coarse spatial detail, no source localisation.



Resolution vs variance

$\Delta f = 1/T$. Shorter segments are noisier; 2 s epochs gave the same AUC (0.812).



ICA

Iterative and random: fixed seed (97), 15 components, blinks found via Fp1/Fp2 as an eye proxy.



Software

scikit-learn's SVM probabilities (Platt scaling) can invert the ranking when the signal is weak.



Metadata

Participant numbers differ between ds004504 and ds006036: per-person age/MMSE links are unreliable.

5.4 Impact Assessment (PO(f))



5.4.1 Societal and Cultural

- EEG costs a fraction of MRI or PET: screening support where imaging is scarce
- Dementia carries stigma: the output means 'refer for assessment', never a label
- Families can plan care earlier



5.4.2 Health and Safety

- Not a diagnosis: 9 of 36 AD missed, 5 of 29 healthy flagged
- Risk of false reassurance or alarm: a clinician must confirm
- EEG is non-invasive and radiation-free



5.4.3 Legal

- Open data (OpenNeuro, CC0 licence), de-identified, used as licensed
- Clinical use would need approval as medical-device software
- Patient EEG must follow data-protection law and consent

Bottom line: a low-cost aid to decide who should be referred for full assessment, never a diagnosis on its own.

5.5 Ethical Issues (PO(h))



Honest analysis

The main result and the robustness criteria were fixed before we saw any result; every result is reported, not only the best.



Reporting our own mistakes

Our first explanation of two below-chance SVMs was tested, rejected and corrected openly (it was the probability step).



EEG is personal data

We could match the same people across two 'anonymous' datasets by their EEG: EEG must be treated as identifiable.



Responsible communication

We reported the ID mismatch to the dataset authors, make no clinical claims, and credit every source.

6.1 Individual Contribution of Each Member

Member	ID	Main contribution
Md. Ashique Al Arafat Nayeem	2206190	Project lead; feature extraction; classification and validation design; main result; dataset ID-mismatch analysis
Arib Salehin	2206186	Robustness and confound analyses (seeds, epochs, rejection, age and sex); limitations, impact and ethics
H M Hamim Al Muhit	2206169	Artefact removal (reference, ICA, rejection); Welch estimator and its validation; cleaning ablation
Shahriar Nafis Rabbi	2206184	Problem formulation and specifications; filter design (FIR, IIR Butterworth, 50 Hz notch)

All members took part in design reviews, checkpoint runs and the rehearsal of every section.



6.2 Logbook of Project

Day 1

Data and tooling

88 recordings loaded,
manifest checked



checkpoint passed

Day 2

First look

group spectra, topographic
maps



checkpoint passed

Day 3

Filter design

FIR 7253 taps vs IIR order
22; notch



checkpoint passed

Day 4

Preprocessing

reference, ICA, rejection



checkpoint passed

Day 5

Spectral estimation

own Welch = SciPy; all 88
processed



checkpoint passed

Day 6

Features

94 regional features,
ranking



checkpoint passed

Day 7

Alpha reactivity

ds006036; participant-ID
mismatch found



checkpoint passed

Day 8

Classification

nested CV: AUC 0.811,
 $p = 0.005$



checkpoint passed

Day 9

What drives it

families, cleaning ablation,
errors



checkpoint passed

Day 10

Robustness

5 of 5 pre-declared criteria
met



checkpoint passed

7. Future Work (PO(I))



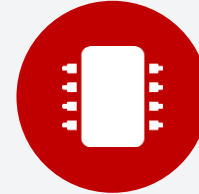
External validation

Test the frozen pipeline on recordings from another hospital and a larger sample.



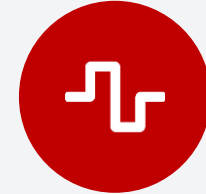
Earlier stages

Mild cognitive impairment, and following the same people over time.



Few-electrode screening

Posterior θ/α needs only a few electrodes: a low-cost headset is plausible.



Richer DSP

Connectivity (coherence, phase-lag index) and source localisation.

Thank you. Questions, and the live demo: one recording in, probability of Alzheimer's out.

8. References

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